

Microcontrollers: Need for Standard Core

Tools and support ecosystem are key drivers for the popularity of a microcontroller architecture. While there are several software tools available for use across different platforms, hardware tools are restricted to only one type of MCU. Considering the cost and time involved in change of a design, isn't there need for a common and consolidated architecture?



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What makes you think of shifting your design from one microcontroller to another? The answer is most likely to be performance and cost. Power consumption, integration, time-to-market, tools and support follow it. But, is it easy to change? Every design engineer, over the course of time, develops an ecosystem and favoritism for a microcontroller architecture that he has been using so far, which makes the shift a little emotional. The change in MCU architecture also incurs a lot of time and money.

You may opt to change the MCU family if your current design is consuming too much power, or the microcontroller architecture does not have the interfaces you require, or

it is too slow. Needless to say, the core architecture of the microcontroller is a key driver for the change. While choosing a new microcontroller, you have to consider all the aforementioned issues and at the same time ensure that the new architecture has right development tools and support. This becomes quite tricky for the designer, as there are so many architectures

from so many vendors available in the market.

According to a report by Frost & Sullivan, the worldwide microcontroller market was \$16 billion in 2007. Surprisingly, as per the report, top two suppliers by revenue shipped more than ten different incompatible architectures of MCUs. Another research from Semico says that the MCU market has more than 40 suppliers providing more than 50 architectures with no architecture holding as much as 5 per cent share in the MCU market. With such a diversity, do we see any commonality between the success of 8051 of yesteryears and today's ARM core?

Core architecture splits the MCU market

There is an ongoing competition between different vendors to offer the

best possible combination of integrated peripherals for dedicated applications. The result is a variety of architectures from different vendors.

Some companies provide their own microcontroller architecture, like Texas Instruments' MSP430, Renesas Electronics' H8 and SuperH, Freescale's Coldfire and Atmel's AVR.

Companies like ARM or MIPS are not microcontroller manufacturers. "We create technology and licence it to companies like STMicroelectronics, NXP, TI and others who build their product families based on our architectures. Hence their products are called ARM-based products," informs Shyam Sadasivan, CPU product manager, ARM.

"We provide intellectual property (IP) of standard processor-based architectures to microcontroller and system-on-chip (SoC) developers," says Ian Anderton, strategic marketing manager, MIPS Technologies.

Although companies like Atmel and TI have their own architectures, these are the licensee of standard cores like 8051 and ARM. "We do not see competition between our own two products, as the business model for both is different. There is a clear distinction from the end applications' viewpoint," says Andreas Eieland, AVR product marketing manager, Atmel.

MCU considerations

Developers need to find out whether the performance of a given microcontroller core matches their product's